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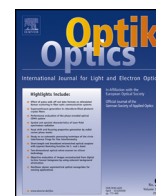


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Modeling and analysis of a highly birefringent chalcogenide photonic crystal fiber



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ABSTRACT

In the present investigation, a highly elliptic chalcogenide core photonic crystal fiber with anisotropic perfectly matched layer (PML) is designed and simulated. Variation of different optical properties of the fiber such as effective refractive index, birefringence, beat length, confinement loss, effective mode area, dispersion and nonlinear coefficient are studied. The unique properties of chalcogenide glass and the structural characteristics of the fiber give high value of nonlinear coefficient, ultra high birefringence, low confinement loss and flat dispersion profile with negative dispersion in the mid-IR wavelength region of the electromagnetic spectrum which is extremely suitable for telecommunication purpose.

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1. Introduction

Photonic crystal fibers (PCFs) are considered as one of the most active fields of current optics research. PCFs are optical fibers that employ a microstructured arrangement of air holes running along the length of the fiber in a background material of different refractive index [1]. The fibers offer many degrees of freedom in their design which is helpful to achieve a variety of peculiar optical properties, which make them attractive for a wide range of applications. A range of novel properties are exhibited in PCFs compared to conventional fibers due to the strong wavelength dependency of the effective refractive index and the inherently large design flexibility of the PCFs. Such properties include endlessly single-mode fibers, extremely nonlinear fibers, large mode area fibers, high birefringence, fibers with anomalous dispersion and many others [2,3].

Birefringence is usually an undesirable property of optical fibers. A high level of birefringence is essential in many sensing applications and in applications where light is expected to maintain a linear polarization state [4]. Birefringent PCFs can be realized easily compared to conventional optical fibers because of higher refractive index contrast between the core and the cladding in contrast of conventional fibers. By destroying the symmetry of the fiber structure and increasing the effective index difference between the two orthogonal polarization modes, high birefringence can be achieved [5]. The structural symmetry can be destroyed either by altering the air hole sizes near the core area or by distorting the shape of

the air holes usually elliptical air holes [6,7]. This causes an increase in the difference between the effective indexes of the orthogonal polarization modes. But this high birefringence is obtained at the cost of poor confinement of light within the fiber since most of the energy is in the cladding. For many applications it is essential to design PCFs that exhibit high birefringence, low confinement losses, small effective area and high nonlinearity simultaneously in a wide wavelength range. In broadband communications systems, fiber dispersion and confinement loss play very important roles.

High index nonsilica based glass fibers have high nonlinearity and low loss in mid-infrared wavelength transmission [8,9]. Chalcogenide glasses contain at least one of the atoms of Se, S or Te in their composition with the addition of As, Ge, Sb and Ga. It can form stable nonlinear media. Fibers fabricated from these glasses are useful for transmission of high power IR light because of their large transparency in the infrared region. Compared to other soft glasses such as fluoride and tellurite glass, chalcogenide glass has a higher nonlinear index and larger refractive index [10]. Chalcogenide glasses possess high third order Kerr nonlinearity up to 1000× higher than silica glass which makes them useful for nonlinear signal processing such as nonlinear switching in optical communication systems. Chalcogenide glasses are highly photo-sensitive with photo-induced refractive index change [11,12].

In this investigation, a highly elliptical chalcogenide photonic crystal fiber is modeled. It is important to study various optical characteristics of the fiber. So in order to understand the fiber behavior, it is numerically investigated and analyzed using finite element method (FEM). Because of large refractive index contrast between the core and the cladding, the transmitting electromagnetic field exhibits high confinement in the core of the fiber. This

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strong confinement leads to high value of nonlinearity without much guiding loss. This proposed PCF demonstrates flattened dispersion along with high nonlinearity.

2. Theory

Compared to standard fiber technology, where the light is guided using solid glasses with different refractive indices, several new properties can be realized using PCF technology. These peculiarities define the applications for which it can be used.

Effective refractive index is defined as a number that quantifies the phase delay per unit length in a waveguide relative to the phase delay in vacuum. In photonic crystal fiber, the refractive index of the cladding is actually an effective refractive index resulting from the introduction of air holes around the core. The effective indices of both fundamental polarization modes are calculated using the finite element method (FEM) which is highly suited for the analysis of our fiber structure. The chromatic dispersion, the effective area and the nonlinear coefficient are then deduced from the determination of the effective indices.

Birefringence or double refraction is the decomposition of a ray of light into two rays named as ordinary ray and extraordinary ray when it passes through certain materials, such as calcite crystals or liquid crystals, depending on the polarization of the light. This effect can occur only if the structure of the material is anisotropic i.e., directionally dependent. The birefringence magnitude is then defined

$$\Delta n = n_e - n_o \quad (1)$$

where n_o and n_e are the refractive indices for polarizations perpendicular (ordinary) and parallel (extraordinary) to the axis of anisotropy, respectively. In a birefringent medium, when two waves with different linear polarization states propagate it, their phases will rotate differently. The phase relation between both waves is restored after integer multiples of polarization beat length or polarization beat distance. This is given by

$$L_b [\text{m}] = \frac{2\pi}{\Delta\beta} = \frac{\lambda}{\Delta n_{\text{eff}}} \quad (2)$$

The confinement loss of PCF is a measure of the amount of power leakage from its core to the cladding. As the confinement loss decreases, the mode area also decreases. But due to the finite number of air holes present in the fiber structure, all the PCF guided modes are leaky [13]. It is found using Eq. (3).

$$L = 8.686k_0 \text{Im}(n_{\text{eff}}) * 10^3 [\text{dB/km}] \quad (3)$$

where $\text{Im}(n_{\text{eff}})$ is the imaginary part of the effective refractive index and $k_0 = 2\pi/\lambda$ is the wave number in the free space.

Effective mode area is a quantitative measure of area that a fiber mode effectively covers in transverse dimensions. The effective area is a very important parameter when studying optical nonlinearities in PCF. Optical nonlinearities depend on the power density inside the device. Therefore, the higher the effective area, the lower will be the effect of nonlinearities for a fixed power. The effective mode area (A_{eff}) is given by

$$A_{\text{eff}} [\mu\text{m}^2] = \frac{\left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |E(x, y)|^2 dx dy \right]^2}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |E(x, y)|^4 dx dy} \quad (4)$$

where E is the magnitude of electric field component.

Dispersion is one of the most important parameter of optical fibers since it can strongly affect the performances of communication systems and fiber-optic nonlinear devices. The dependence of the speed of light on its wavelength is referred to as dispersion. Due to dispersion, a signal will be delayed or distorted. The sum of the

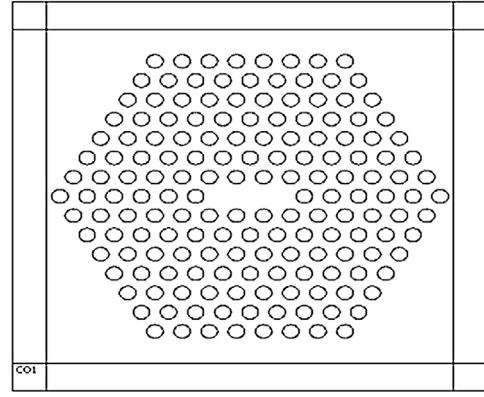


Fig. 1. Schematic of the proposed hexagonal structure PCF with PML. $\Delta = 0.8 \mu\text{m}$, $d = 0.6^* \Delta$.

material and waveguide dispersion constitutes the total dispersion or chromatic dispersion. The waveguide dispersion of fundamental mode can be defined as

$$D_w(\lambda) = \left(\frac{-\lambda}{c} \right) \left(\frac{d^2 n_{\text{eff}}}{d\lambda^2} \right) \quad (5)$$

Here n_{eff} is found by using the Sellmeier's formula. The material dispersion of chalcogenide glass is given by

$$D_w(\lambda) = \left(\frac{-\lambda}{c} \right) \left(\frac{d^2 n(\lambda)}{d\lambda^2} \right) \quad (6)$$

The total dispersion of PCFs is

$$D(\lambda) = D_m(\lambda) + D_w(\lambda) [\text{ps/nm km}] \quad (7)$$

Owing to the wavelength scale structure and capability of tight field confinement, PCFs can offer very small mode area A_{eff} and enhance the effective nonlinearity. The nonlinear parameter γ is defined as

$$\gamma = \frac{2\pi n_2}{\lambda A_{\text{eff}}} [\text{W}^{-1} \text{m}^{-1}] \quad (8)$$

where n_2 is the nonlinear refractive index of chalcogenide glass, λ is the wavelength of the light and A_{eff} is the effective mode area of PCF. High nonlinear coefficient is of great advantage for number of applications among which supercontinuum generation is most important.

3. Design of proposed highly birefringent photonic crystal fiber

The proposed fiber consists of six rings of hexagonal lattice circular air holes in the cladding region which is bounded by anisotropic perfectly matched layer (PML). PML is an additional domain that absorbs the incident radiation without producing reflections. Six rings of air holes are used to reduce the confinement loss. All air holes are circular with diameter d and pitch Δ . The asymmetric core of the fiber is made by removing the central three air holes from the regular hexagonal pattern. The air filling fraction is 0.6 and the pitch is $0.8 \mu\text{m}$. The refractive index of Te-enriched bulk chalcogenide glass $\text{Ge}_{20}\text{As}_{20}\text{Se}_{14}\text{Te}_{46}$ is approximately 3.3 [14]. The schematic of the proposed structure of PCF is shown in Fig. 1.

4. Results and discussion

The COMSOL simulated output of proposed fiber is shown in Fig. 2. The mode field is highly confined in the core of the PCF for

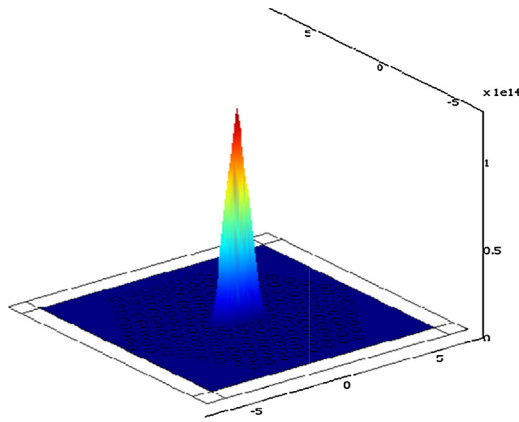


Fig. 2. 3D view of the proposed fiber after simulation.

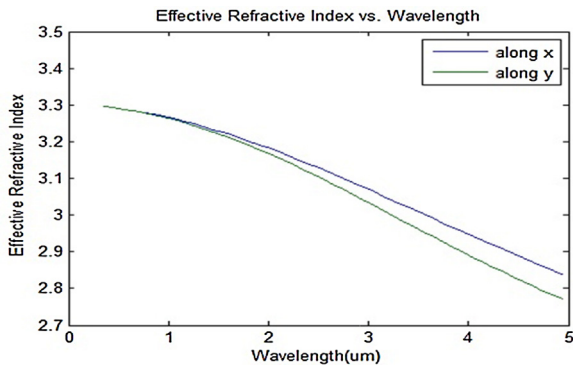


Fig. 3. Variation of effective refractive index (n_{eff}) with wavelength (λ).

a wide range of wavelengths and exhibits endlessly single mode nature. When the wavelength of light is smaller than the core diameter of PCF, the propagating mode will be fully confined in the core. When the wavelength is greater than the core diameter, mode starts leaking in to cladding, thereby increasing the loss.

The anisotropic nature of the fiber results in two different refractive-index profiles along the x and y axes. The variation of effective refractive indices along x and y directions is shown in Fig. 3. The effective refractive index along both directions decreases with increase in wavelength. This change in refractive index gives birefringence.

Birefringence in optical fibers can be obtained by geometrical, stress or a combination of both effects. This fiber gives high birefringence due to geometrical anisotropy. The variation of birefringence of the designed PCF with wavelength is shown in Fig. 4. The birefringence increases with increase in wavelength. The birefringence of the order of 10^{-2} is obtained for a wide range of wavelengths in this simple structure without using any elliptical air holes which leads to high confinement loss. In telecommunication and sensor

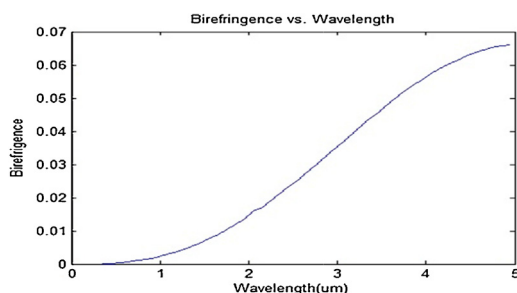


Fig. 4. Variation of birefringence with wavelength (λ).

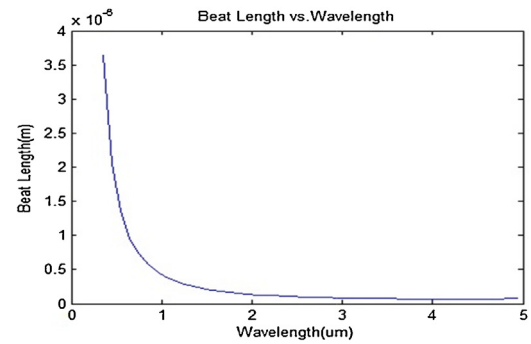


Fig. 5. Variation of beat length with wavelength (λ).

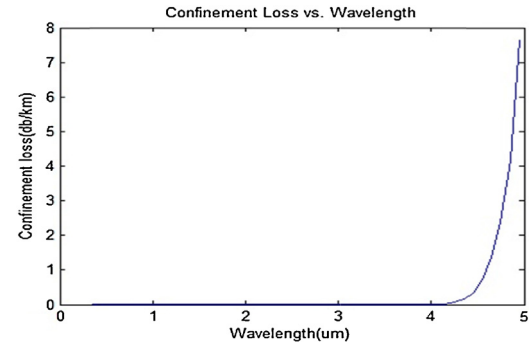


Fig. 6. Variation of confinement loss with wavelength.

application field, the presence of inherent and induced birefringence is crucial.

Birefringence and beat length are inversely proportional. So the low value of beat length which is in μm range is due to high value of birefringence in this fiber. The variation of beat length with wavelength is shown in Fig. 5. The fiber has high polarization-holding capacity because of low value of beat length. As a result, the fiber can act as polarization maintaining fiber.

Then we considered the confinement loss of the fiber. The variation of confinement loss (db/km) with wavelength is depicted in Fig. 6. Even though this fiber exhibits high birefringence, the confinement loss is very less. The confinement loss increases with increase in wavelength. The confinement loss is less than 0.1 db/km up to 4.3 μm . In all the three telecommunication windows, the confinement loss is of the order of 10^{-10} . This is due to increased number of rings, presence of PML structure and the absence of elliptical air holes.

Another important property that we investigated is dispersion. The chromatic dispersion profile can be easily controlled by varying the hole diameter and the hole-to-hole spacing. Controllability of chromatic dispersion in PCFs is a very important problem for practical applications to optical communication systems, dispersion compensation, and nonlinear optics [15]. The variation of dispersion with wavelength is shown in Fig. 7. This fiber has two zero dispersion wavelengths at 3.08 μm and 4.02 μm . Also shows very high value of negative dispersion in lower wavelengths and low higher order dispersion between the two zero dispersion wavelengths. So this fiber has a flat dispersion profile with relatively constant group delay dispersion over some wavelength range. Such fibers are important for data transmission with wavelength division multiplexing and for adiabatic soliton compression. A low and flat dispersion profile is important for the maximization of the super continuum generation in fiber since it can decrease the temporal walk-off effect during the spectral boarding process. Because of high negative dispersion in all the three telecommunication windows, this can be a dispersion compensation fiber also.

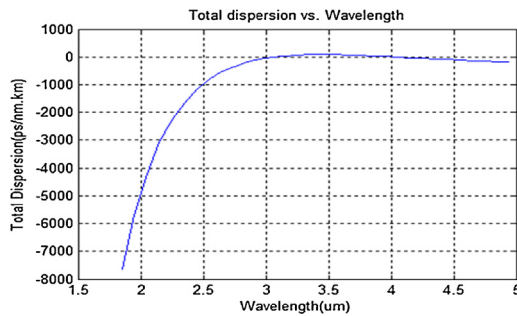


Fig. 7. Variation of total dispersion with wavelength.

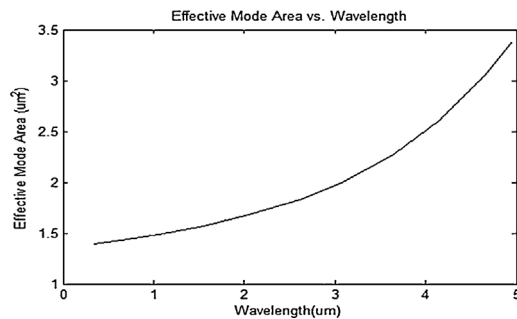


Fig. 8. Variation of effective mode area with wavelength.

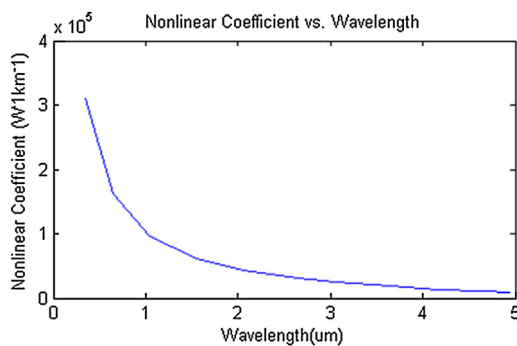


Fig. 9. Variation of nonlinear coefficient with wavelength.

The effective mode area is a very important parameter when studying optical nonlinearities in PCF. Higher the mode area, the lower will be the effect of nonlinearities. The variation of effective mode area and nonlinear coefficient with wavelength is shown in Figs. 8 and 9, respectively. The effective mode area increases with wavelength while nonlinearity decreases with wavelength. The nonlinear coefficient is extracted from the effective area and higher index difference leads to higher nonlinear coefficient. The mode area is $1.57 \mu\text{m}^2$ at $1.55 \mu\text{m}$.

The nonlinear refractive index of chalcogenide glass $\text{Ge}_{20}\text{As}_{20}\text{Se}_{14}\text{Te}_{46}$ is not reported anywhere. The nonlinear refractive index of As_2Se_3 is 2.3×10^{-17} [16]. The value of $\text{Ge}_{20}\text{As}_{20}\text{Se}_{14}\text{Te}_{46}$ would be definitely more than this. Even by using this value, this fiber gives high nonlinearity of $311,000 \text{ W}^{-1} \text{ km}^{-1}$ at $0.4 \mu\text{m}$, $24,900 \text{ W}^{-1} \text{ km}^{-1}$ and $13,900 \text{ W}^{-1} \text{ km}^{-1}$ at first and second ZDW, respectively. The magnification of effective nonlinearity offers totally new possibilities in the field of all-optical signal

processing, since it enables the realization of compact nonlinear devices with low power requirements [17].

5. Conclusion

This fiber is a highly birefringent and nonlinear chalcogenide asymmetric solid core photonic crystal fiber with low confinement loss, mode area, beat length and flat dispersion profile. This fiber finds extensive application in the fields of polarization controllers, sensors and optical communication in mid-infrared wavelengths. In terms of passive optical functions, chalcogenide photonic crystal fibers are useful for power delivery, nulling interferometry for space-based applications and pollutant detection using optical sensors. Nonlinear fibers with such a dispersion profile can provide the way for a range of new broadband tunable devices like tunable optical parametric amplifiers, wavelength converters, regenerators and all-optical demultiplexers and also for super continuum generation.

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